Supplementary Materials

Social Presence in Community Responses to VTuber Learning Content:
An Exploratory Study Across Roles and Media

This document provides supporting information that could not be included in the manuscript: the full list of questionnaire items (Section S1), the list of analyzed videos and theme distributions by video and by region (Section S2), and per-video proportions for the comparison between live streams and on-demand videos (Section S3). Channel names, video titles, and video IDs are not included to protect the privacy of creators and commenters; videos are referred to by the same codes as in the manuscript.

S1 Supporting Questionnaire

The questionnaire in this study was shown in Indonesian. Part 1 recorded demographic data and VTuber content consumption patterns: age (under 18, 18–24, 25–34, and over 34 years), gender, viewing frequency (daily, several times a week, occasionally, rarely or almost never), the types of VTuber content previously watched (multiple answers allowed: learn/study with me, co-study/co-working, material or education delivered by a VTuber, or none), and usual behavior when watching learning or productivity content (only watching, or actively writing in the chat or comment section).

Part 2 presented the same 15 statements (Table S1) three times, once for each content category, in the following order: (1) *Learn/Study with me*, reported in the manuscript as **learning together (LT)**; (2) *Co-study / Co-working*, reported as **co-stream (CS)**; and (3) *Penyampaian materi/edukasi oleh VTuber* (material or education delivered by a VTuber), reported as **learning material (MLV)**, covering both **live streams** and **on-demand videos**. Each statement was rated on a five-point Likert scale (1 = strongly disagree, 5 = strongly agree). The questionnaire ended with one question asking which content type was most helpful for learning or productivity.

Table S1. Questionnaire statements (original Indonesian and English translation)

Code	Aspect	Original (Indonesian)	English translation
SP1	Social Presence	Saya merasakan kehadiran VTuber seolah menemani saya.	I feel the VTuber's presence as if they were accompanying me.
SP2	Social Presence	Saya merasa terhubung secara sosial dengan VTuber maupun penonton lain saat menonton konten ini.	I feel socially connected with the VTuber and other viewers when watching this content.
SP3	Social Presence	Konten ini membuat saya merasa tidak sedang belajar/bekerja sendirian.	This content makes me feel that I am not studying/working alone.
FS1	Focus Support	Kehadiran VTuber membantu saya tetap fokus pada tugas saya sendiri.	The VTuber's presence helps me stay focused on my own tasks.
FS2	Focus Support	Menonton konten ini mendorong saya untuk tidak menunda pekerjaan.	Watching this content encourages me not to procrastinate.
FS3	Focus Support	Saya lebih mampu mempertahankan konsentrasi ketika "ditemani" VTuber dibanding mengerjakan sendiri.	I can maintain concentration better when "accompanied" by a VTuber than when working alone.
MO1	Motivation	Menonton konten ini meningkatkan motivasi saya untuk belajar/bekerja.	Watching this content increases my motivation to study/work.
MO2	Motivation	Konten ini membuat saya bersemangat menyelesaikan aktivitas saya.	This content makes me eager to complete my activities.
MO3	Motivation	Saya terdorong belajar/bekerja lebih giat setelah menonton konten ini.	I feel encouraged to study/work harder after watching this content.
EV1	Educational Value	Saya memperoleh informasi atau wawasan yang bermanfaat dari konten ini.	I gain useful information or insights from this content.
EV2	Educational Value	Konten ini membantu saya memahami materi/topik yang dibahas.	This content helps me understand the material/topic discussed.
EV3	Educational Value	Konten jenis ini cocok digunakan sebagai sarana belajar.	This type of content is suitable as a learning medium.
DI1^{R}	Distraction	Kehadiran/gaya VTuber justru mengalihkan perhatian saya dari tugas yang seharusnya dikerjakan.	The VTuber's presence/style distracts me from the tasks I should be doing.

Code	Aspect	Original (Indonesian)	English translation
DI2^{R}	Distraction	Saya lebih memperhatikan avatar/VTuber-nya daripada materi atau tugas saya.	I pay more attention to the avatar/VTuber than to the material or my task.
DI3^{R}	Distraction	Konten ini lebih terasa menghibur daripada membantu saya belajar/bekerja.	This content feels more entertaining than helpful for studying/working.

^{R}Reversed: reverse-coded before analysis (higher scores indicate less distraction).

S2 Analyzed Videos and Theme Distributions

Table S2.1 lists the 12 analyzed videos. Each of the four data cells contains one Indonesian or Malaysian, one global English-language, and one Japanese video.

Table S2.1. Analyzed videos

Code	Lang.	Topic	Uploaded	Duration	Views	Raw / Clean
LT1	MS	Biology exam papers	2026-05-07	3:35:58	3,351	3,886 / 3,584
LT2	EN	Japanese via photos	2025-12-04	3:56:22	560	675 / 575
LT3	JP	Aurora/earth science	2024-05-11	3:12:03	3,082	1,520 / 1,481
CS1	JP	Medical exam study	2026-06-17	11:01:43	4,923	547 / 527
CS2	ID	Python programming	2025-12-04	3:36:37	257	550 / 538
CS3	EN	Video editing	2026-06-15	2:48:46	677	354 / 234
ML1	ID	Health myths Q&A	2026-03-28	2:01:54	485	2,226 / 2,175
ML2	EN	Japanese Q&A	2026-06-05	3:16:50	923	734 / 681
ML3	JP	Science event report	c. 2023^{R}	1:26:56	$\approx$1,000^{R}	814 / 705
MV1	ID	Earthquake explainer	2024-09-24	0:01:41	58,230	374 / 366
MV2	EN	Car engines	2025-04-09	0:07:00	133,633	740 / 733
MV3	JP	Art/national treasures	2024-08-27	0:06:35	367,625	500 / 495

LT = learning together; CS = co-stream; ML = learning material, live stream; MV = learning material, on-demand video. MS/ID = Malaysia/Indonesia, EN = global, JP = Japan. Views as of September 2026. Raw / Clean = number of comments before and after noise removal. ^{R}Set to private by its creator after data collection; upload year and views (as of June 2026) are approximate.

One hundred comments were sampled from each video, so the counts in Table S2.2 are also percentages of that video's comments, and each category subtotal comprises 300 comments, corresponding to Fig. 2 in the manuscript.

Table S2.2. Theme distribution by video (number of comments out of 100 per video)

Video	Lang.	SP	EV	MO	FS	DI	AA*	HQ*	Meta*	Total
LT1	MS	38	38	14	7	0	3	0	0	100
LT2	EN	41	34	13	5	3	4	0	0	100
LT3	JP	34	45	4	14	1	1	0	1	100
LT subtotal		*113*	*117*	*31*	*26*	*4*	*8*	*0*	*1*	*300*
CS1	JP	46	12	5	28	2	5	0	2	100
CS2	ID	36	44	2	13	4	1	0	0	100
CS3	EN	52	21	8	11	0	6	0	2	100
CS subtotal		*134*	*77*	*15*	*52*	*6*	*12*	*0*	*4*	*300*
ML1	ID	58	13	1	0	6	1	14	7	100
ML2	EN	36	52	10	0	0	2	0	0	100
ML3	JP	40	43	15	1	0	1	0	0	100
ML subtotal		*134*	*108*	*26*	*1*	*6*	*4*	*14*	*7*	*300*
MV1	ID	18	37	19	2	1	21	0	2	100
MV2	EN	30	39	9	8	1	13	0	0	100
MV3	JP	17	62	9	4	1	7	0	0	100
MV subtotal		*65*	*138*	*37*	*14*	*3*	*41*	*0*	*2*	*300*
All videos		**446**	**440**	**109**	**93**	**19**	**65**	**14**	**14**	**1200**

SP = Social Presence; EV = Educational Value; MO = Motivation; FS = Focus Support; DI = Distraction; AA = Avatar Appeal; HQ = Humorous Q&A; Meta = Meta-stream. *Inductive themes. LT = learning together; CS = co-stream; ML = learning material (live stream); MV = learning material (on-demand video). MS/ID = Malaysia/Indonesia, EN = global English-speaking, JP = Japanese.

Table S2.3 groups the same data by the region of the viewing community. Each region contains four videos, one from each category (400 comments).

Table S2.3. Theme distribution by region, n (%)

Theme	Malaysia/Indonesia (MS/ID)	Global (EN)	Japan (JP)
Social Presence	150 (37.5)	159 (39.8)	137 (34.2)
Educational Value	132 (33.0)	146 (36.5)	162 (40.5)
Motivation	36 (9.0)	40 (10.0)	33 (8.2)
Focus Support	22 (5.5)	24 (6.0)	47 (11.8)
Distraction	11 (2.8)	4 (1.0)	4 (1.0)
Avatar Appeal*	26 (6.5)	25 (6.2)	14 (3.5)
Humorous Q&A*	14 (3.5)	0 (0.0)	0 (0.0)
Meta-stream*	9 (2.2)	2 (0.5)	3 (0.8)

Each region contains 400 comments. *Inductive themes. MS/ID = Malaysia/Indonesia, EN = global, JP = Japan.

S3 Live Streams Versus On-Demand Videos

Table S3.1 lists the per-video counts for the instructor role (300 live-stream comments and 300 on-demand comments in total). Because each video contributed 100 comments, the counts in Table S3.1 are also percentages.

Table S3.1. Per-video counts in the instructor role (out of 100 comments per video)

	Live streams			On-demand videos		
Theme	ML1	ML2	ML3	MV1	MV2	MV3
Social Presence	58	36	40	18	30	17
Educational Value	13	52	43	37	39	62
Motivation	1	10	15	19	9	9
Focus Support	0	0	1	2	8	4
Distraction	6	0	0	1	1	1
Avatar Appeal*	1	2	1	21	13	7
Humorous Q&A*	14	0	0	0	0	0
Meta-stream*	7	0	0	2	0	0

*Inductive themes. ML3 was set to private by its creator after data collection; its comments are reported only in aggregate.

The comparison between live streams and on-demand videos was computed per video rather than per comment, because comments from the same video tend to resemble one another (they share the same topic, audience, and atmosphere). The comparison is therefore between three live streams and three on-demand videos. For each theme, the difference in proportions (live minus on-demand) was estimated by bootstrapping, that is, by resampling videos with replacement within each medium (10,000 resamples, seed 2026), and the 2.5th and 97.5th percentiles were used as the 95% confidence interval. An exact permutation test on the six video-level proportions was also computed; with three videos per medium there are only 20 possible allocations, so the smallest attainable two-sided p is .10.

Table S3.2. Live versus on-demand differences in the instructor role

Theme	**Live** n (%)	**On-demand** n (%)	**Difference** (pp)	**95% CI** (bootstrap)	**Permutation** p
Social Presence	134 (44.7)	65 (21.7)	23.0	[11.3, 36.0]	0.10
Educational Value	108 (36.0)	138 (46.0)	−10.0	[−33.0, 11.3]	0.60
Avatar Appeal*	4 (1.3)	41 (13.7)	−12.3	[−19.7, −6.0]	0.10
Focus Support	1 (0.3)	14 (4.7)	−4.3	[−7.7, −1.7]	0.10
Motivation	26 (8.7)	37 (12.3)	−3.7	[−11.7, 4.3]	0.60
Distraction	6 (2.0)	3 (1.0)	1.0	[−1.0, 5.0]	1.00

Each medium contains 300 comments from three videos. pp = percentage points (live minus on-demand). An interval that excludes zero indicates a difference supported by the bootstrap. *Inductive theme. Humorous Q&A (live only, ML1) and Meta-stream are omitted because of small counts.